ASSIGNMENT 2

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1.0 Introduction

This report applies clustering to group customers by how much they spend.
The Wholesale Customers dataset was used.
K-Means and Hierarchical Clustering was applied.
The goal was to find groups of customers who spend in similar ways.
These groups help businesses plan marketing, improve sales, and manage stock.

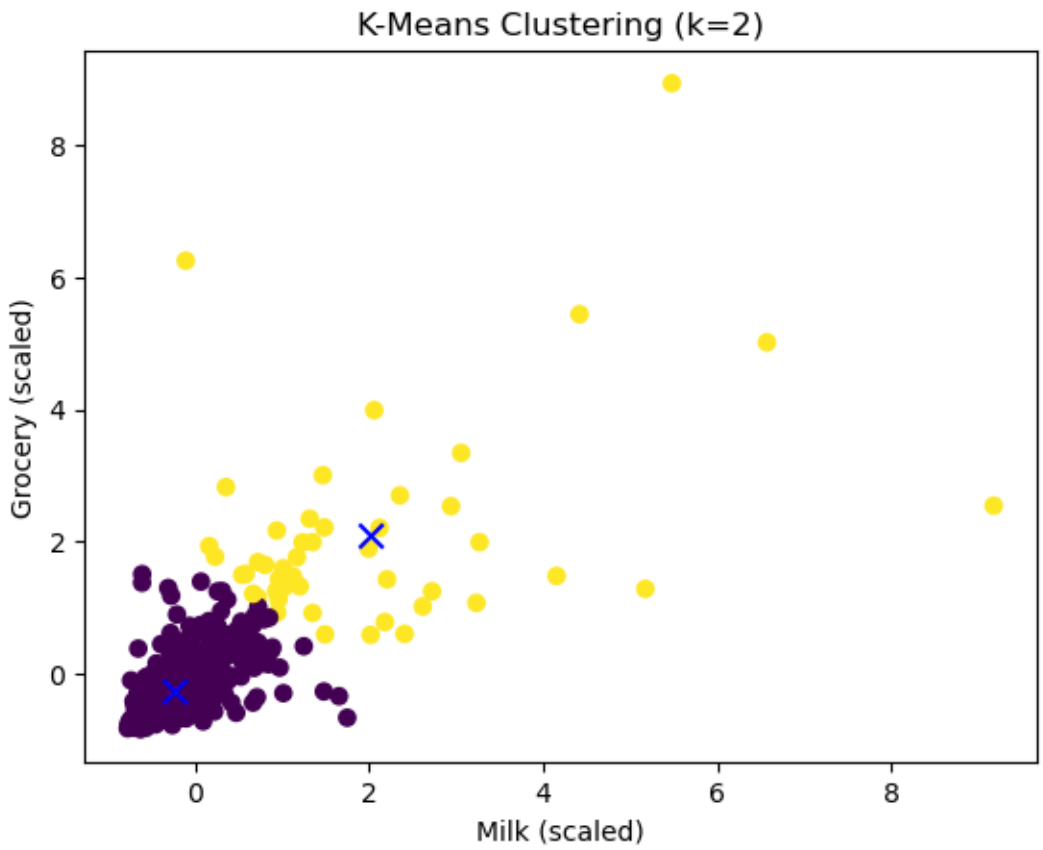
2.0 Data Preprocessing

The dataset includes spending on Fresh, Milk, Grocery, Frozen, Detergents_Paper, and Delicatessen.
Each value is in monetary units.
Channel and Region are also given but are not numeric.
Only numeric features were used for clustering.
Clustering depends on distance. Large numbers can dominate smaller ones.
To fix this, features were standardized.
Standardization changes all values to have mean 0 and standard deviation 1.
This keeps each feature equal in influence.

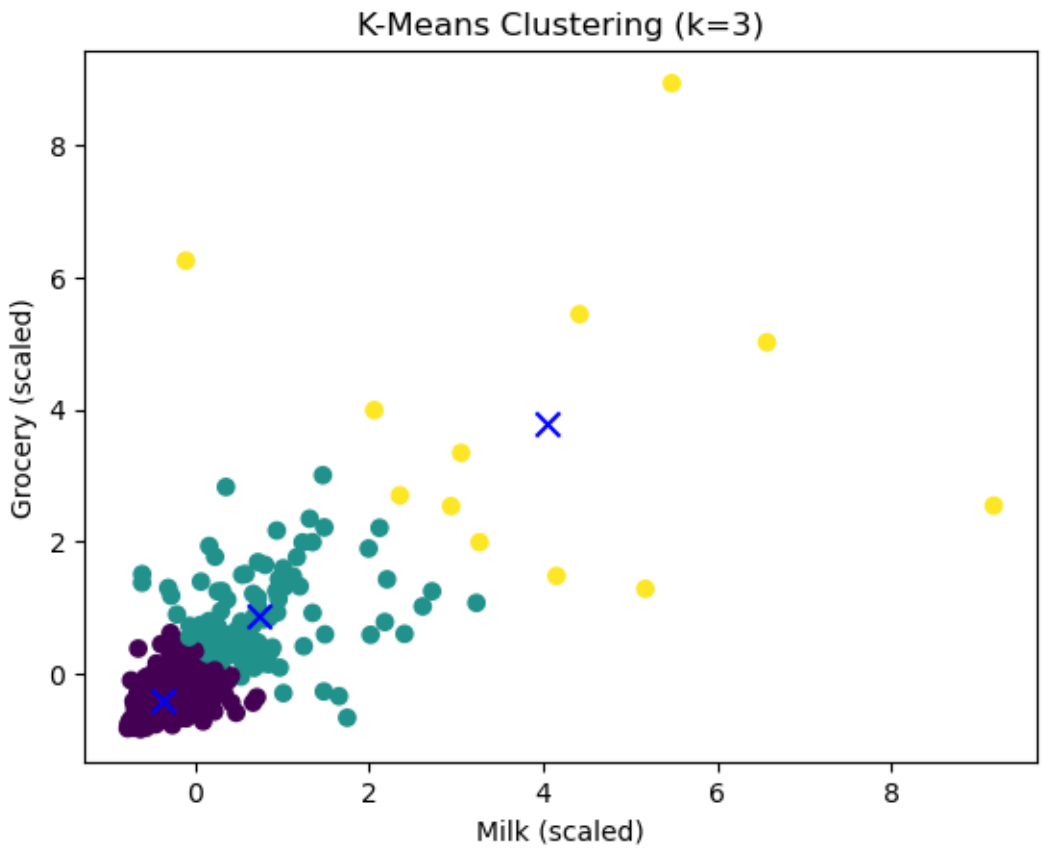
3.0 K-Means Clustering

3.0.1 Observing the Number of Clusters

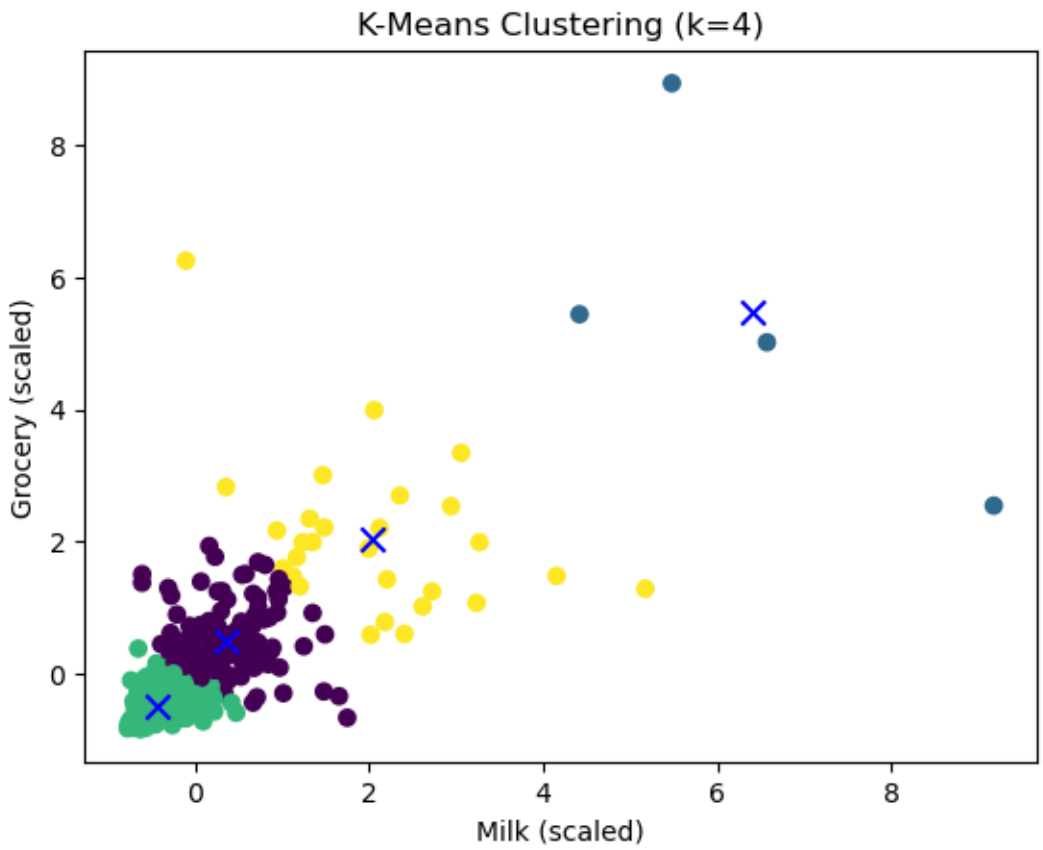
K-Means was tested for k values from 2 to 7.
The features Milk and Grocery were used for plots.



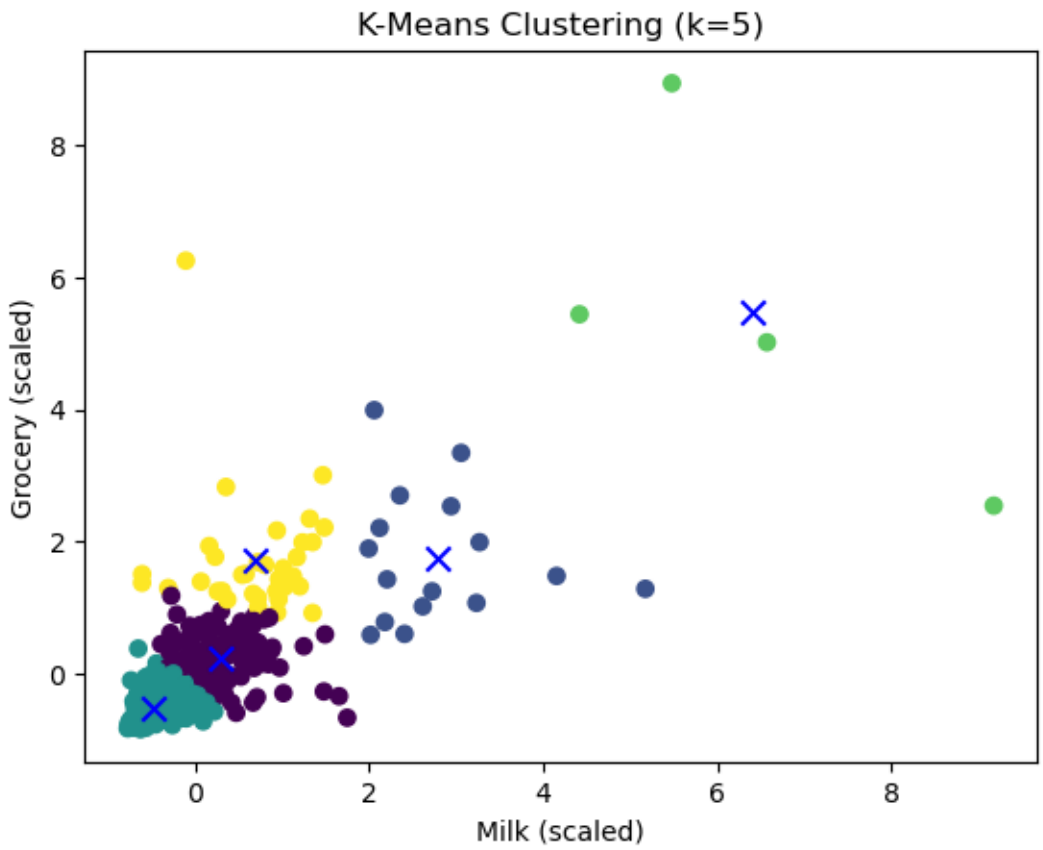
At k = 2, two main groups appeared.
The separation looked clear and even.



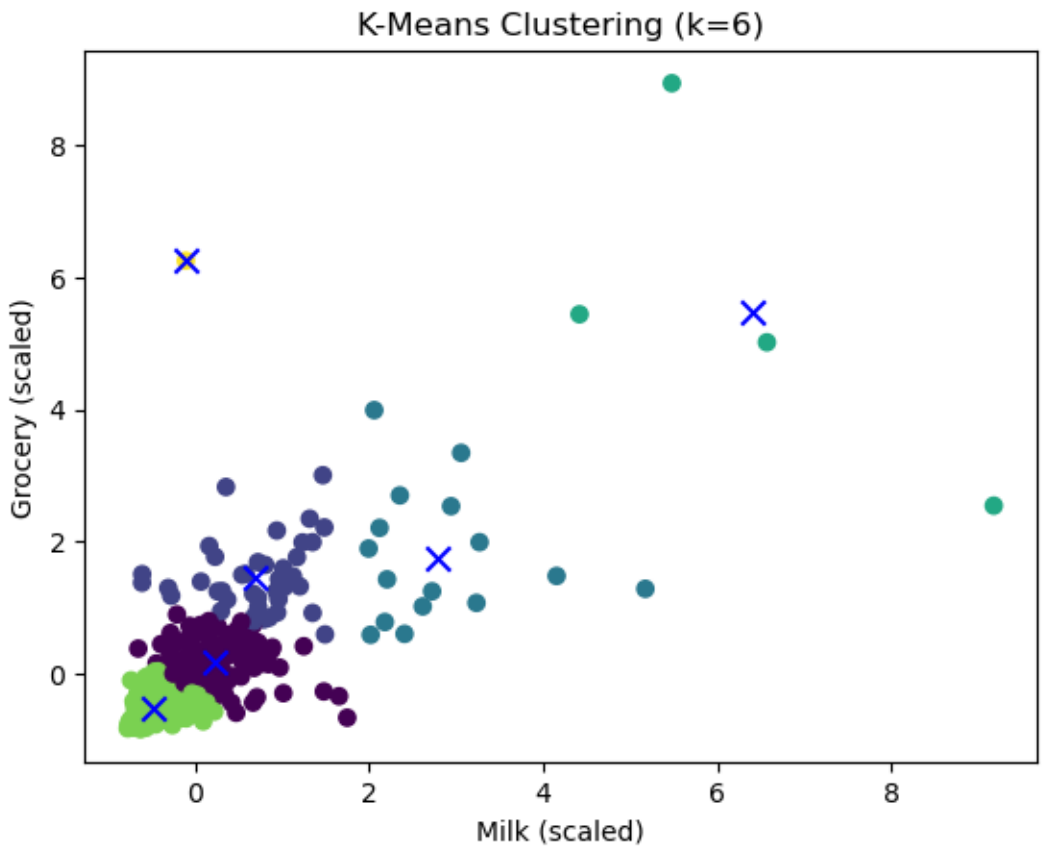
At k = 3, a third group appeared in between.
The layout looked smoother.



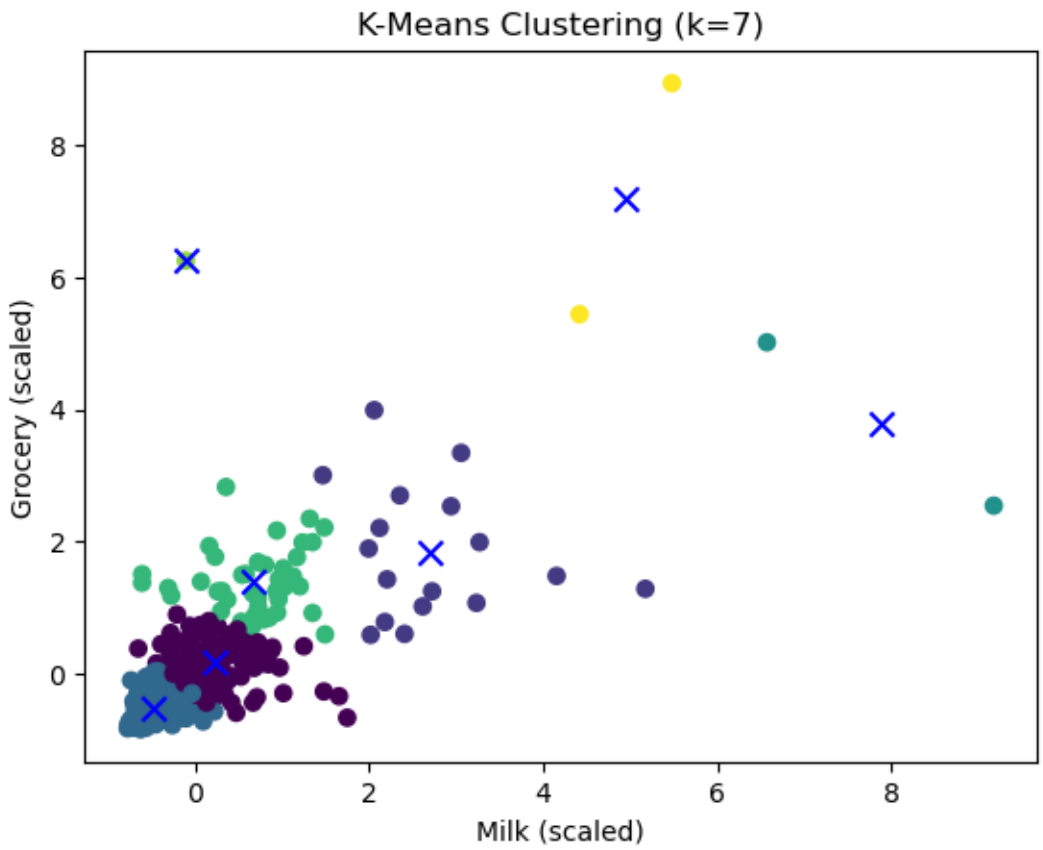
At k = 4, the clusters looked more refined.
One cluster at the corner had points far from its centre.
This showed that the points there did not fit closely together.



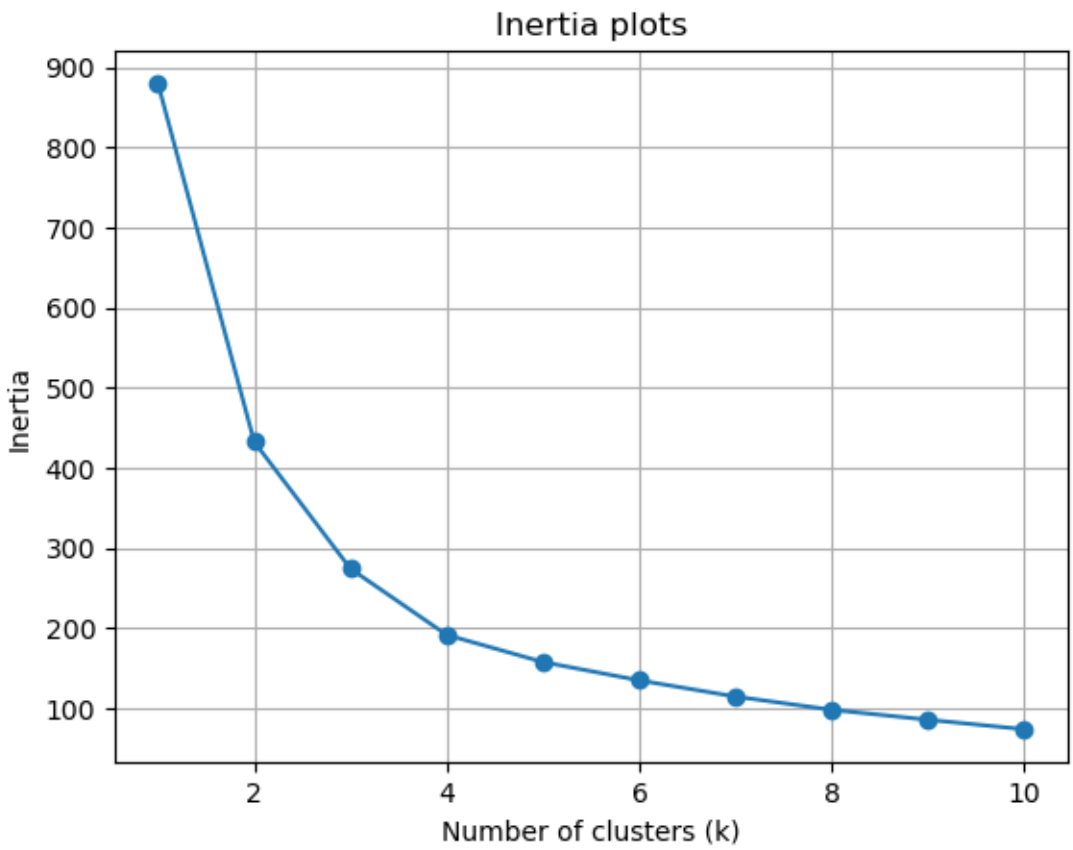
At k = 5, the clusters started to split more.
The boundaries did not improve much.



At k = 6, one cluster had only a single point.
This showed that the clusters were starting to lose meaning.



At k = 7, some clusters were large and loose.
Points were spread far from their centres.

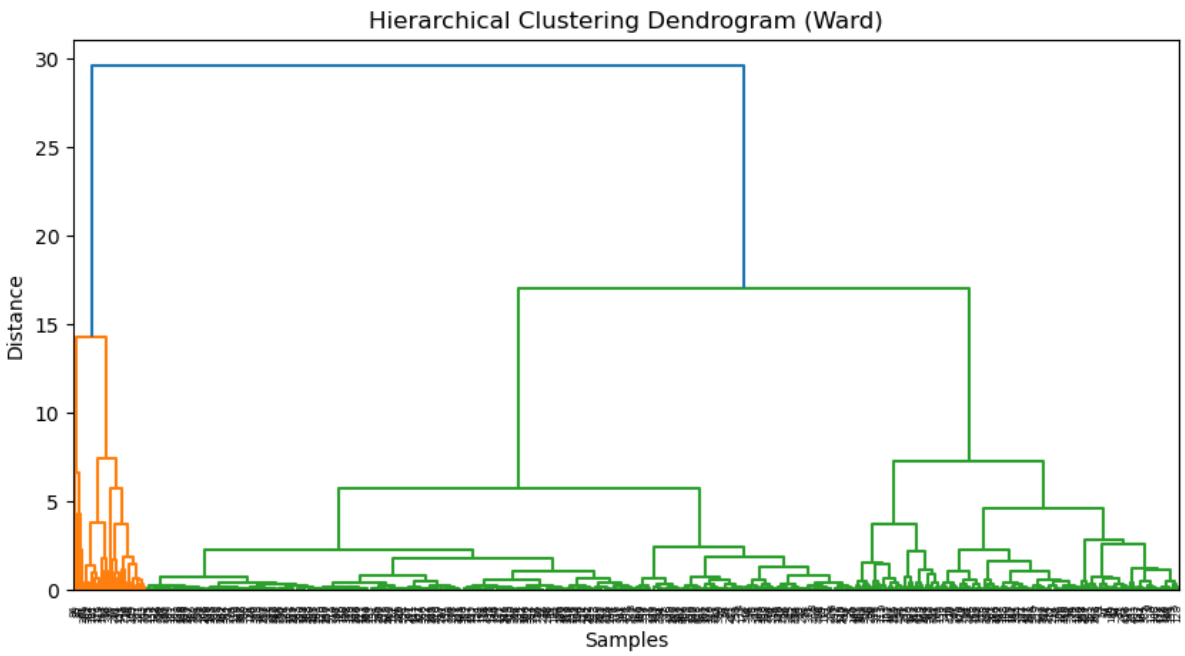


The inertia dropped sharply from k = 1 to k = 2.
After that, it flattened.
This means most of the improvement happened by k = 2.
From k = 4 onwards, loose and small clusters started to appear.
Adding more clusters did not help.
Based on the curve and plots, k = 2 was chosen as best.

4.0 Hierarchical Clustering

4.0.1 Dendrogram and Optimal Cluster Selection

Hierarchical clustering was applied using Ward's linkage.
The dendrogram showed how clusters merged step by step.



Two large gaps were seen — one between 14 to 28 and another between 7 to 14.
These gaps showed where very different clusters merged.
The cut was made around linkage distance 11.
This gave three big clusters and one small cluster on the left.
The smaller merges below this point joined similar points.
Above it, the merges joined groups that were very different.
The cut at 11 gave a clear four-cluster solution.

5.0 Cluster Interpretation and Comparison

5.0.1 Strengths and Weaknesses

K-Means Results

Ten clusters were formed.
The sizes were 203, 24, 3, 106, 10, 1, 9, 1, 49, and 34.
Most points were in the larger clusters.
A few clusters were very small.
Those tiny clusters were not useful.
The silhouette score was 0.449.
This shows that the clusters were only somewhat separate.
Centroid values for Milk and Grocery ranged from −0.56 and −0.59 up to

9.18 and 6.46.
This shows that some customers spend very little while others spend a lot.

Hierarchical Results

When cut into three clusters, the sizes were 29, 129, and 282.
This was more balanced.
The silhouette score was 0.541, which was higher.
The average Milk and Grocery values were 2 479 / 3 301, 8 515 / 12 620, and 25 962 / 32 404.
This shows clear spending levels: low, medium, and high.

K-Means Strengths

Fast on large data.
Simple to understand and easy to plot.

K-Means Weaknesses

The number of clusters must be chosen first.
It can make very small or uneven clusters.

Hierarchical Strengths

No need to pre-set the number of clusters.
The dendrogram shows how data merges.
It produced clean and balanced groups.

Hierarchical Weaknesses

It takes longer to run.
It can be affected by outliers.

6.0 Comparison of Unsupervised and Supervised Methods

Clustering is unsupervised.
To compare, a Support Vector Machine (SVM) was trained using the Region label.
The accuracy was about 0.72.

SVM Classification Results

Metric	Precision	Recall	F1-Score	Support
Region 1	0.00	0.00	0.00	23

Region 2	0.00	0.00	0.00	14
Region 3	0.72	1.00	0.84	95
Overall Accuracy				0.72 (≈ 72 %)

Region 3 was predicted correctly.
Regions 1 and 2 were not.
Most records belonged to Region 3.
This made the model focus on that class.

When K-Means clusters were compared to Region labels, accuracy was also about 0.72.
Most clusters matched Region 3.
One small cluster matched Region 2.
This means both methods found a similar structure.
SVM used labels to learn boundaries.
Clustering found groups naturally.
Both reached similar results, but in different ways.

Unsupervised Learning (K-Means)

Finds hidden patterns.
Good for exploring unlabeled data.
Clusters may not always match real categories.

Supervised Learning (SVM)

Gives high accuracy when labels exist.
Cannot find new or unseen groups.

7.0 Business Insights and Recommendations

The clusters show how customers differ in how much they spend.
K-Means showed two main groups.
Customers with high Milk and Grocery values spend more.
Those with small or negative values spend less.

The hierarchical model gave three spending levels: low, medium, and high.
Their average spending values were 2 479 / 3 301, 8 515 / 12 620, and 25 962 / 32 404.
This gives a clear picture of spending patterns.

High-spending customers can be given loyalty programs or premium offers.
Medium spenders can be targeted with discounts and cross-selling.
Low spenders can receive promotions or bundles to increase buying.

Both clustering and SVM showed about 72 % accuracy when compared to Region.
This means region influences spending.
Regional promotions and stock plans can be made based on this.

Clustering helps businesses make decisions using data.
It supports better marketing, planning, and customer management.

8.0 Conclusion

Two clustering methods were applied to customer data.
K-Means found two main groups.
Hierarchical clustering produced four clusters but with three main ones.
Both methods showed that spending patterns can be grouped clearly.
SVM confirmed that the clusters reflect real regional behaviour.
Clustering gives insight into how customers behave.
This helps businesses plan and make better decisions.